TECNOLÓGICO NACIONAL DE MÉXICO

Instituto Tecnológico de Ciudad Guzmán

Course. Power coditioners for renewable energy sources

TOPIC. Global Warming Awareness Campaing

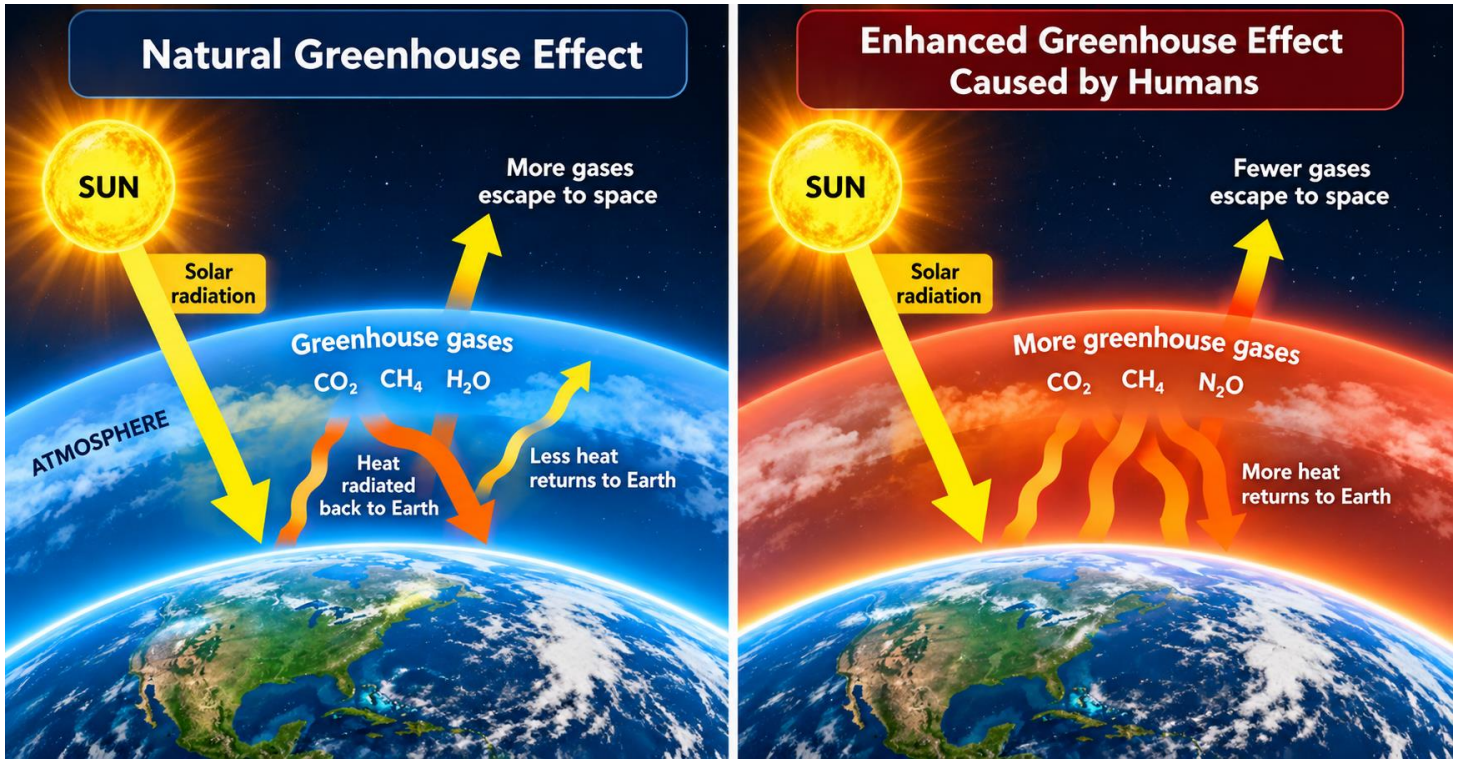
CAMPING. Cultura Energética y Conciencia Climática

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Introduction to the Problem

Global warming is the **progressive increase in the Earth's average temperature**, mainly caused by rising concentrations of greenhouse gases in the atmosphere. These gases trap some of the heat that would otherwise escape into space, causing changes in the planet's temperature.



VOA. (n.d.). *Diagram of the natural greenhouse effect and the enhanced greenhouse effect caused by human activities.* Image adapted and digitally enhanced using AI.

The greenhouse effect is a natural process in which certain gases in the atmosphere trap some of the Sun's heat, helping to keep the Earth warm.

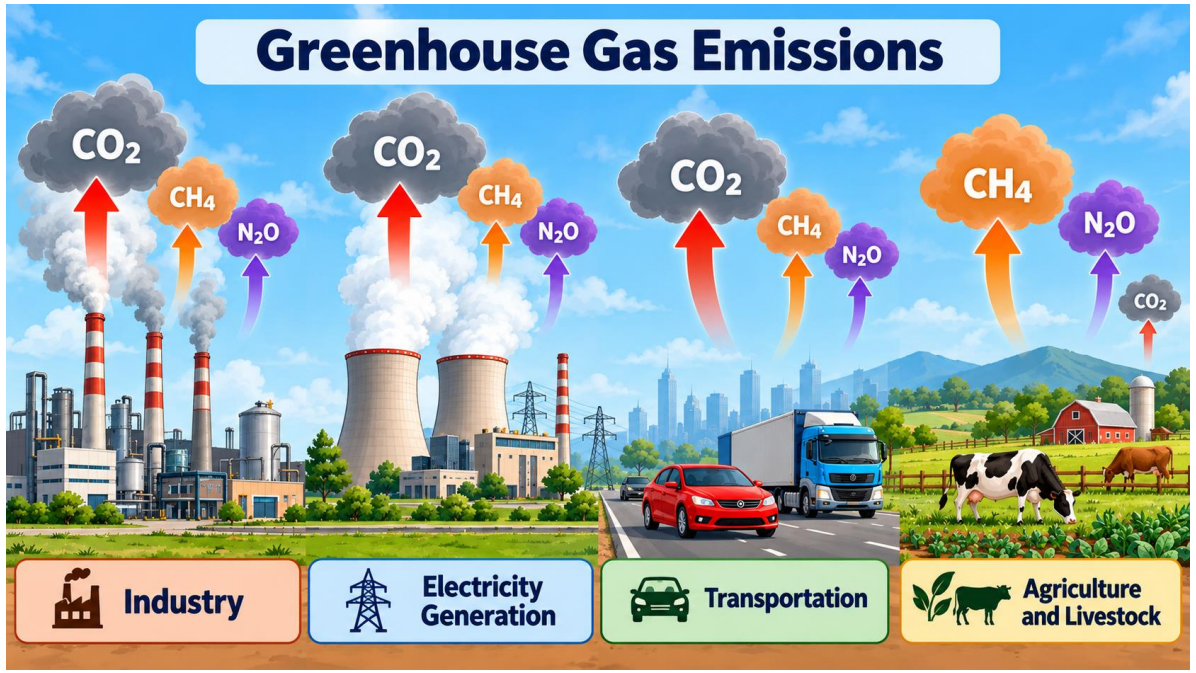
The problem begins when there are too many greenhouse gases, because more heat is trapped, causing the planet's temperature to rise.

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The Industrial Revolution: The Main Problem

Since the Industrial Revolution, humanity has massively increased the use of **coal, oil, and natural gas** to power factories, trains, cars, ships, generate electricity, and manufacture materials. Burning these fossil fuels releases large amounts of **CO_2**, one of the main greenhouse gases. In addition, industrialization also

- **The production of cement and steel**, which releases CO_2.
- **Deforestation**, as forests were cleared for cities, agriculture, and industry.
- **Intensive livestock farming and agriculture**, which produce methane (CH_4).
- **The use of fertilizers**, which releases nitrous oxide (N_2O).



OpenAI. (2026). *Greenhouse Gas Emissions* [AI-generated image].

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Implementation Methodology

The campaign will be implemented at the Tecnológico de Ciudad Guzmán through an academic strategy based on student projects. Its development will be integrated into research workshops, research-related courses, and professional residency projects, with the aim of encouraging students to propose real-world solutions to global warming.

Methodology Stages

1. Campaign Planning
2. Selection of Project Topics
3. Development of Academic Projects
4. Dissemination within the Institute
5. Evaluation of Results



OpenAI. (2026). *Stages of the implementation methodology for an environmental campaign.*

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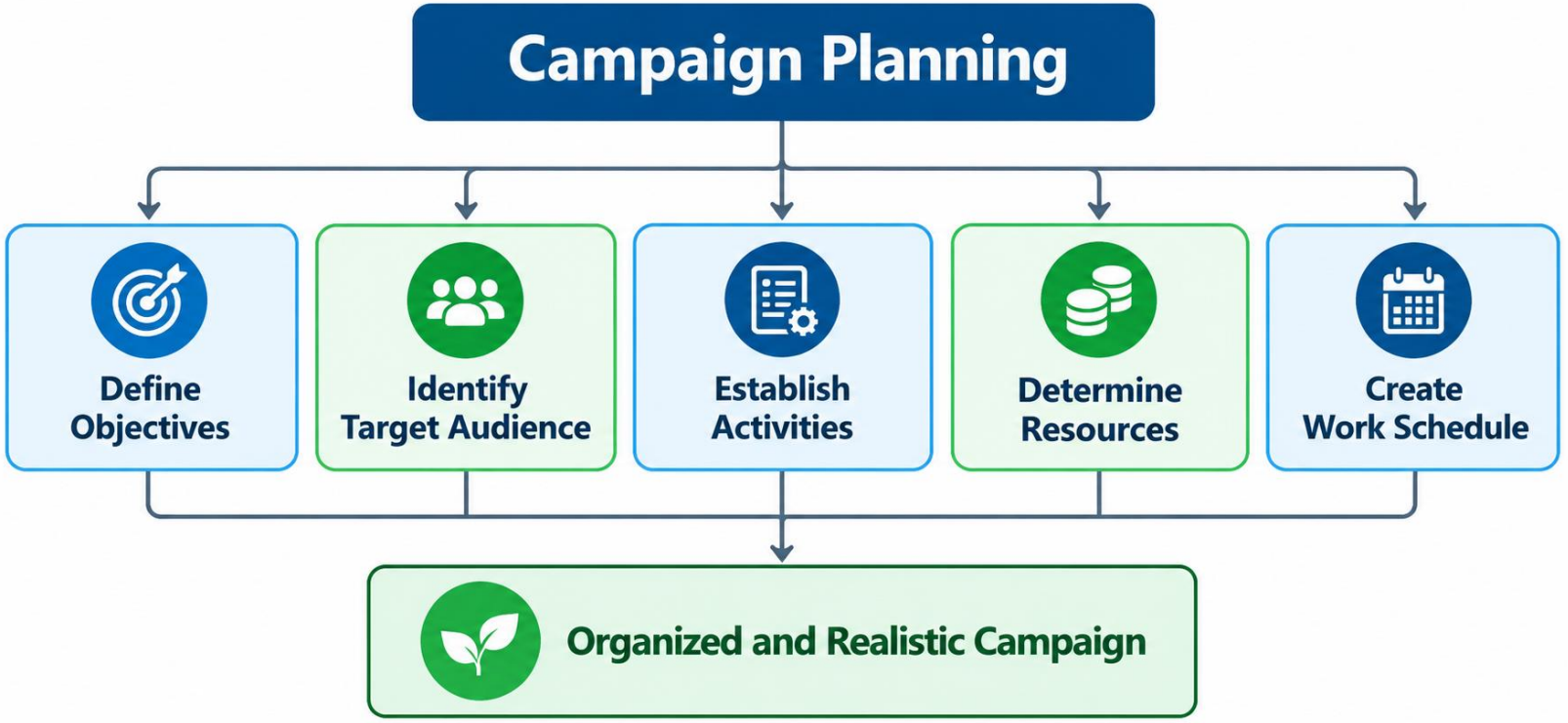
Implementation Stages

1. Campaign Planning

The **objectives, target audience, main activities, required resources, and work schedule** will be defined. This stage will allow the campaign to be organized in a **clear and realistic manner** within the Tecnológico de Ciudad Guzmán.



OpenAI. (2026). *Stage 1 of the implementation methodology.*



OpenAI. (2026). *Campaign planning diagram* [AI-generated image].

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1. Campaign Planning



OpenAI. (2026). Source: Own elaboration with the support of OpenAI about campaing planning.

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Implementation Stages

2. Selection of Project Topics

Students will select areas of work related to renewable energy, energy efficiency, emissions reduction, environmental monitoring, or sustainable practices. The aim is to guide the projects toward real-world and applicable solutions.



OpenAI. (2026). *Stage 2 of the implementation methodology.*



OpenAI. (2026). *Research lines for sustainable projects* [AI-generated image].

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Implementation Stages

3. Development of Academic Projects

The selected topics will be integrated into research workshops, research-related courses, and professional residency projects. At this stage, students will be able to design proposals, prototypes, campaigns, or technological systems.



OpenAI. (2026). *Stage 3 of the implementation methodology.*

Year	Authors	Study focus	Educational application	Relation to the proposal
2019	**Lozano et al.** *(Int. J. Sustainability in Higher Education)*	University–industry collaboration for sustainability	Collaborative projects with real stakeholders	Linking students with companies to develop sustainable solutions
2021	**Santos et al.** *(Int. J. Sustainability in Higher Education)*	Project-based learning to develop sustainability competencies	Student projects focused on real problems	Integration of selected themes into academic projects and workshops
2024	**Zhou et al.** *(Cleaner Production Letters)*	Knowledge exchange for sustainability (systematic review)	Student projects with real impact	Development of solutions with environmental and social impact
2025	**Chen et al.** *(Discover Education)*	Sustainable product design in engineering (DT-CDIO-RA framework)	Design and development of prototypes	Creation of technological solutions for sustainability
2026	**El Youssfi, BenFilali & El Hammoumi** *(Discover Education)*	Sustainability in engineering education (project-based learning)	Projects on renewable energy, energy efficiency and smart technologies	Integration in academic courses and research workshops
2026	**Rahman et al.** *(Higher Education, Skills and Work-based Learning)*	Integration of sustainability in final engineering projects	Capstone projects and professional internships	Application in professional residencies and final academic projects

Table 1. Previously developed works on academic projects focused on sustainability.
Source: Own elaboration based on works reported in the scientific literature (2019–2026).

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Implementation Methodology

4. Dissemination of Results

The developed projects will be shared within the Tecnológico de Ciudad Guzmán through exhibitions, posters, infographics, presentations, institutional social media, or demonstrations. This will allow the campaign to reach a wider audience among students.



OpenAI. (2026). *Stage 4 of the implementation methodology.*

Conference / Event	Organizing Institution	Location	Main Topics	Type of Participation
CIREDII *Ibero-American Congress on Intelligent Systems and Industrial Applications*	CIREDII Congreso Iberoamericano de Sistemas Inteligentes y Aplicaciones Industriales	**Mexico** (various venues)	Automation and control Renewable energy IoT and intelligent systems Sustainability and environment	Oral presentations, posters and scientific articles
CNEER National Congress of Renewable Energy Students	CNEER Congreso Nacional de Estudiantes de Energías Renovables	**UNAM** (Institute of Renewable Energy) Mexico City, Mexico	Renewable energy Energy efficiency IoT and smart grids Climate change and sustainability	Student projects, oral presentations and posters
CIERMMI International Congress on Renewable Energy, Mechatronics, Computing and Education	CIERMMI Congreso Internacional Interdisciplinario de Energías Renovables, Mecatrónica, Informática y Educación	**Universidad Veracruzana** Boca del Río, Mexico	Renewable energy Mechatronics and automation Information technologies Education and sustainability	Oral presentations, posters and scientific articles
CIDEI International Congress of Engineering	CIDEI Congreso Internacional de Ingeniería	**Universidad Autónoma de Coahuila** Saltillo, Mexico	Intelligent systems Energy and climate change Automation and control Sustainability	Oral presentations, posters and technological projects
CIENERGIA UG International Congress on Energy	CIENERGIA UG Congreso Internacional de Energía	**Universidad de Guanajuato** Guanajuato, Mexico	Energy and water Automation and Industry 4.0 Sustainable technologies Environment and climate change	Scientific articles, research projects and student presentations
CCTI Scientific-Technological Congress of Engineering	FES CUAUTITLÁN UNAM	**FES Cuautitlán – UNAM** State of Mexico, Mexico	Engineering and technology development IoT and automation Sustainability and environment Applied projects	Oral presentations, posters and student projects

Table 2. Conferences where related academic and technological projects have been presented. Source: Own elaboration based on official information from the conferences (2019–2026).

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Implementation Methodology

5. Impact Evaluation

Finally, the results of the campaign will be analyzed by considering student participation, the projects developed, and the level of environmental awareness achieved. This evaluation will help identify areas for improvement in future activities.



OpenAI. (2026). *Stage 5 of the implementation methodology.*

University	Company / Organization	Student project	Action against global warming	Main results / Impact
University of Washington (USA)	amazon	Design and analysis of solar energy solutions for Amazon facilities	Replace conventional electricity with renewable energy	Reduction of greenhouse gas emissions and lower carbon footprint
Nottingham Trent University (UK)	MILLWARD	Carbon footprint reduction strategy for the company	Identify and reduce emission sources in company operations	Recommendations to reduce emissions and achieve sustainability goals
University College London (UK)	MOTT MACDONALD	Sustainability assessment in construction projects	Reduce embodied carbon and improve sustainable construction practices	Strategies to lower carbon emissions and environmental impact in infrastructure projects
Purdue University (USA)	PEPSICO	Feasibility study of biodiesel and hydrogen for industrial operations	Replace fossil fuels with low-carbon alternative fuels	Roadmap for decarbonization and potential reduction of industrial emissions
Delft University of Technology (Netherlands)	Shell	Optimization of offshore wind energy systems	Increase the use of renewable energy in the energy sector	Technical and economic recommendations to expand clean energy generation
Universidad de Chile (Chile)	CODELCO	Water and energy efficiency in mining operations	Reduce energy consumption and associated emissions in mining processes	Lower energy use, reduced emissions and more sustainable mining operations

Table 3. Examples of student projects developed in collaboration with companies and focused on global warming mitigation. Source: Own elaboration based on information reported by universities and participating companies (2019–2026).

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Expected Results

The campaign is expected not only to increase awareness of global warming, but also to motivate students to transform this knowledge into technological and sustainable projects, preferably linked to real-world challenges faced by companies and organizations.

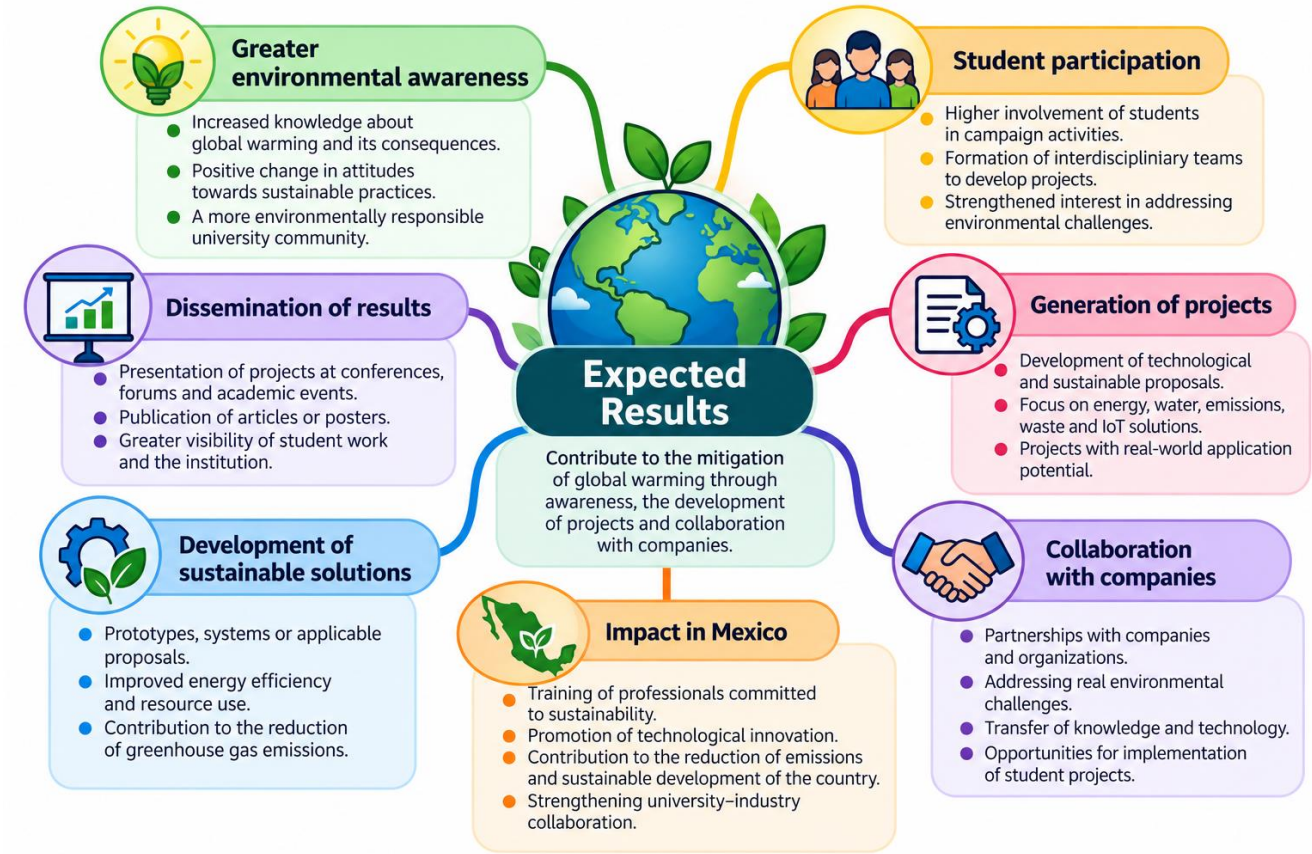


Table 4. Expected results of the global warming awareness campaign.
Source: Own elaboration with the support of OpenAI (2026).

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1. National Aeronautics and Space Administration. (n.d.). *Global warming*. NASA Science. https://science.nasa.gov/earth/climate-change/global-warming/
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